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Joint Radar Emitter and Mode Clustering Using a Transformer Encoder Architecture

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Abstract

This thesis investigates joint clustering of radar emitters and their operating modes from intercepted pulse descriptor word (PDW) streams using a transformer encoder architecture.

Keywords: transformer, deep clustering, electronic intelligence, radar emitter identification, mode recognition, self-supervised learning.

Sammanfattning

Detta examensarbete undersöker gemensam klustring av radarsändare och deras operativa moder från avlyssnade pulsbeskrivande ord (PDW) med hjälp av en transformerbaserad encoderarkitektur.

Nyckelord: transformer, djup klustring, signalspaning, radarsändaridentifiering, modigenkänning, självövervakad inlärning.

Acknowledgments

I would like to thank my supervisor and examiner at KTH, and the host organization for their support throughout this thesis project.

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Markus Swegmark

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List of Acronyms

ELINT electronic intelligence

ESM electronic support measures

EW electronic warfare

PDW pulse descriptor word

TOA time of arrival

CHAPTER 1

Introduction

1.1 Background and Motivation

Start with story about Ukraine and Information operations (IO).

Electronic warfare (EW) is becoming increasingly important in modern warfare as technological advancements continues in radar technology, and the advancement of computer algorithms, such as machine learning, which can be superior to humans at some delimited tasks.

One of the main objectives of electronic warfare is to control the electromagnetic spectrum. Identifying radars in your surroundings is a crucial part of this. In the subset of EW called electronic support measure (ESM), where one passively listens to the electromagnetic environment to infer and deduce useful information about it, especially information about other agents in it, including mapping which radars are present and information about them.

In modern electronic warfare, so called agile radars, which can be anything from missiles to airplanes to stationary radars, are becoming more and more sophisticated. When emitting a signal for detection, they are intentionally varying the characteristics of the signal, like frequency, pulse width, amplitude, pulse repetition interval to hide their presence in an increasingly more unpredictable way and in expanding ranges, enabled by modern hardware and evolving algorithms (including machine learning).

This leads to a “cat and mouse”-like situation, where the agile radar agent wants to either scan an area, or lock into a target, sometimes being the entity with the ESM, or something else, without being detected themselves, or in other ways try and confuse the ESM.

Apart from having different algorithms for varying the pulse signal, the agile radars also send out different signals depending on the objective, for example if it’s scanning it might have a radar rotating around an axis, leading to an oscillating received signal, while if the radar wants to track a single object it can send a more direct signals, leading to a constant signal in the naïve case. Usually, the algorithms for varying the signal characteristic are pretty simple, maybe it changes frequency in a set sequence or random etc, and keeps it like this. However, this can will make it hard to identify and also more advanced algorithms of course exists. This algorithm configuration for varying the signal characteristics combined is what we refer to a mode of the radar (this includes if it’s searching or locking on). If we know the mode of a radar, we basically know how the signal will look like (plus minus some noise and interference etc), and also it tells us about the agent sending out the signal if it’s searching or locking in, and also can be used to distinguish between emitters and tell us of how advanced the radar is and perhaps even the goal of the agent. Because of this, it would be ideal to identify not only what signals comes from what radar, but also what modes they are using. Note that two radars can have same modes, but also change the modes. We might need to make some limitations and assume that one radar stays in the same mode. Our input data also give us angle of incoming signal which should enable us to filter or give support for classifying different radars from positioning.

In the field of ES we usually call it radar classification (RC) if we want to classify what type of emitter the pulses are coming from. E.i SA-6 radar, drone XXY, etc. While if we want to say exactly what instance it is, that is also distiguish the number of emitters from each emitter as well, e.i SA-6 radar 3, drone XXY 1 and 3, we call that specific radar identification (SRI). So you can say that RC is a superset of SR, and since it provied more informaiton it is more advantageous - however the marginal strategic gain from information of SRI compared to RC can be discuess while the marginal cost might be discussed. The marginal gains is we know the number of emitters exactly, and the ID hopefully, however this problem is sometimes harder and might have lower accurary compared to pure RC, thus decreaseing the Marginal gains. The marginal cost is however that out model might get more complicated, and depending on our model, it could also decrease the accuracy of the classification problem - we will not allow this to happened.

1.2 Problem Statement

Modern radar systems are evolving rapidly. They can change their transmission behavior on the fly (often called "agile" radars), which makes it increasingly difficult for traditional, rule-based methods to classify and identify them reliably. In Electronic Warfare (EW), Electronic Support Measures (ESM) systems passively listen to the electromagnetic environment and aim to detect, characterize, and identify radar emitters in order to enhance situational awareness and protection. Radars emit short bursts of energy known as pulses. Different radars tend to produce characteristic patterns across pulse features such as carrier frequency, pulse width, time of arrival, and angle of arrival. By analyzing these patterns, it is possible to infer how many emitters are present and which emitters they are. However, overlapping signals, agile behavior, noise, and dynamic scenarios make this a complex pattern recognition problem.

This thesis addresses the end-to-end ML classification problem of determining, from recorded radar pulses, both the number of radar emitters present in a flight simulation and their identities. The work will use labeled, simulator-generated ESM data provided by Saab, in which each pulse is described by features such as frequency, pulse width, time of arrival (ToA), and angle of arrival (AoA). The problem is formulated as a supervised learning task, with the goal of training models that map pulse observations to emitter identities. For this thesis you will both design an AI model but also analyze the simulated data. To understand how good the model is it is also required to understand the difficulty of the data it is evaluated on. You will create metrics for measuring dataset complexity and then compare how your model performs on data of varying complexity according to your metrics.

1.3 Research Questions

The thesis addresses the following research questions:

RQ1 Placeholder research question.

RQ2 Placeholder research question.

1.4 Objectives and Scope

Placeholder paragraph.

1.5 Contributions

The main contributions of this thesis are:

- Placeholder contribution.
- Placeholder contribution.

1.6 Ethical and Societal Considerations

Placeholder paragraph.

1.7 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 introduces the necessary background on radar signal processing and deep learning. Chapter 3 surveys related work. Chapter 4 describes the proposed method. Chapter 5 presents experiments and results. Chapter 6 discusses the findings. Chapter 7 concludes and outlines future work.

CHAPTER 2

Background

2.1 Radar Systems and Passive Reception

Radio is the art of communicating using electromagnetic waves, traditionally done, but not limited to, radio waves. Discovered in theory in the electromagnetic revolution in the 1800s by the Maxwell and others, and made reality in the end of that centrury by Guglielmo Marconi who in 1891 who transmitted the first morse code over a kilometer using radio waves from his device. xt

Radar, which is an abbreviation for radio detection and range, is a form of echolocalisation, using radio waves. It was discovered in the early 1900s, and it's development was greatly boosted due to it's utility detecting plans and warships in world war 2.

2.2 Radar Emitters and Operating Modes

Placeholder paragraph.

2.3 Pulse Descriptor Words

Placeholder paragraph.

2.3.1 EW Signal Processing Pipeline

In EW and ESM contexts, the path from the electromagnetic environment to downstream reasoning is often described as a *signal processing pipeline*. At a high level, wideband receivers capture emissions, detectors mark pulse events, and estimators extract per-pulse parameters such as carrier frequency, pulse width, and time of arrival. Deinterleaving then groups pulses into coherent trains and emits PDWs (or equivalent feature vectors) as the interface to higher-level ELINT tasks, including emitter and mode analysis. Placeholder paragraph.

2.4 Machine Learning Fundamentals

Placeholder paragraph.

2.5 Clustering Methods

Placeholder paragraph.

2.6 Deep Representation Learning

Placeholder paragraph.

2.7 The Transformer Encoder

Placeholder paragraph.

CHAPTER 3

Related Work

3.1 Classical Radar Emitter Identification

Placeholder paragraph.

3.2 Deep Learning for Radar Signal Analysis

Placeholder paragraph.

3.3 Deep Clustering

Placeholder paragraph.

3.4 Transformers for Time Series and Sets

Placeholder paragraph.

3.5 Joint and Multi-Task Clustering

Placeholder paragraph.

CHAPTER 4

Method

4.1 Problem Formulation

This chapter studies unsupervised structure recovery from a time-ordered stream of PDWs produced by an ESM or deinterleaving front end, as outlined in Section 2.3.1. The goal is twofold: associate each pulse with a physical *emitter instance* and, within that instance, with an *operating mode*, without prior knowledge of how many emitters or modes appear in the data. Both quantities are latent because labels are scarce in ELINT settings and because novel emitters and modes must be accommodated.

Observed input

Let a sequence of N pulses be represented by feature vectors

$$\boldsymbol{X} = (\boldsymbol{x}_1, \dots, \boldsymbol{x}_N), \qquad \boldsymbol{x}_n \in \mathbb{R}^d, \tag{4.1}$$

where each $\boldsymbol{x}_n$ encodes the measured pulse parameters associated with one PDW (carrier frequency, pulse width, timing, spatial cues, and any auxiliary fields). The sequence may contain irregular sampling, missing pulses, and spurious detections inherited from upstream processing; robustness to such effects is a design constraint rather than part of the formal specification.

Latent emitter and mode assignments

For each index $n \in \{1, \dots, N\}$, let e_n denote the emitter identity responsible for pulse n and let m_n denote the operating mode within that emitter. These variables take values in finite label sets whose cardinalities

$$K_{\text{em}} = |\{e_1, \dots, e_N\}|, \qquad K_{\text{md}} = |\{m_1, \dots, m_N\}| \tag{4.2}$$

are *unknown a priori* and need not match the number of physical platforms in the environment (for example, co-located antennas may map to a single inferred emitter index, or a single platform may split across indices when observations are ambiguous). The pair (e_n, m_n) is not observed at training time; the learning problem is to infer representations from which both clusterings can be recovered reliably.

Desired outputs: dual embeddings and discrete labels

The parametric model considered in this thesis maps each input sequence to two parallel trajectories in Euclidean embedding spaces,

$$\boldsymbol{Z}^{\mathrm{em}} = (\boldsymbol{z}_1^{\mathrm{em}}, \ldots, \boldsymbol{z}_N^{\mathrm{em}}), \quad \boldsymbol{z}_n^{\mathrm{em}} \in \mathbb{R}^p, \qquad \boldsymbol{Z}^{\mathrm{md}} = (\boldsymbol{z}_1^{\mathrm{md}}, \ldots, \boldsymbol{z}_N^{\mathrm{md}}), \quad \boldsymbol{z}_n^{\mathrm{md}} \in \mathbb{R}^q, \tag{4.3}$$

with dimensions p and q fixed by the architecture. Discrete emitter labels $\hat{e}_n$ and mode labels $\hat{m}_n$ are obtained *post hoc* by clustering $\{\boldsymbol{z}_n^{\mathrm{em}}\}_{n=1}^N$ and $\{\boldsymbol{z}_n^{\mathrm{md}}\}_{n=1}^N$, or by reading off prototype assignments when the training objective includes cluster parameters. Thus the immediate trainable targets are the continuous vectors in Equation (4.3); hard partitions are induced indirectly.

The two trajectories serve different roles. Emitter-oriented vectors $\boldsymbol{z}_n^{\mathrm{em}}$ are primarily a vehicle for estimating consistent emitter membership; their main deliverable is a per-pulse emitter index suitable for downstream ELINT workflows such as emitter counting and localisation. Mode-oriented vectors $\boldsymbol{z}_n^{\mathrm{md}}$ additionally aim to summarise mode-dependent waveform behaviour so that operating modes can be inspected, compared, or matched against a library of known modes once such references become available.

Architecturally, both streams are produced from a shared encoder $f_{\boldsymbol{\theta}}$ with task-specific heads (see Section 4.3); in deployment, emitter embeddings may be less critical to retain after discrete $\hat{e}_n$ have been fixed, whereas mode embeddings (or intermediate encoder activations feeding the mode head) are natural inputs to specialised mode classifiers.

Learning objective

The optimisation problem couples representation learning with clustering-friendly losses so that the geometry of $\boldsymbol{Z}^{\mathrm{em}}$ and $\boldsymbol{Z}^{\mathrm{md}}$ aligns with emitter and mode structure, respectively. The concrete loss terms and training protocol are given in Sections 4.4 and 4.5.

4.2 Data Representation and Preprocessing

ELINT data are difficult to obtain at scale, and operational recordings are often sensitive. This work therefore relies on synthetic pulse trains from a proprietary scenario simulator (Saab), which yields controlled scenarios and full supervision of emitter identity and operating mode for evaluation, while the learning objectives remain unsupervised with respect to those labels.

Each scenario is a time-ordered sequence of PDWs together with emitter and mode annotations. Scenarios vary in the number of emitters and in sequence length so that training covers diverse traffic densities and temporal horizons; exact counts, train/validation/test splits, and sampling ranges are reported in Chapter 5.

Windowing and context length

The encoder operates on fixed-length windows because self-attention cost scales quadratically with sequence length. Let L denote the window length (in pulses) fed to the model; experiments use $L = 1024$. Each scenario is partitioned into contiguous windows of length L. If the remainder after the last full window is shorter than L, it is dropped in the present pipeline; padding with an attention mask would retain the tail without changing the architecture and is left as a straightforward extension.

Training windows are extracted with a stride of $L/2$, so consecutive windows overlap by half their length. This increases the number of training pulses seen per scenario and reduces sensitivity to the particular alignment of window boundaries relative to mode transitions. Validation and test windows use stride L (no overlap), so that metrics are not inflated by near-duplicate windows.

Per-feature scaling

Amplitude is already provided on a logarithmic scale (decibels). Continuous fields are transformed before windowing so that statistics are computed on the training split only and then applied to validation and test data. The log-amplitude, carrier frequency, and pulse width are standardised to zero mean and unit variance using the training-set mean and standard deviation of each field. Each TOA is first min–max scaled to $[0, 1]$ using training-set bounds; within every window, all rescaled TOA values are shifted by subtracting the TOA of the first pulse so that the window starts at time zero. This removes absolute clock offset while preserving intra-window timing structure.

Windows are grouped into mini-batches for optimisation; batch size and per-epoch shuffling of windows are stated in Chapter 5.

4.3 Transformer Encoder Architecture

Architectutre used was tranofmer encoders stacked. Without any posivaional encoding since the paper XXhttps://arxiv.org/pdf/2503.13476 sadi so , and we have teh TOA already. We use batch firzt and layer norm. The first 6 layers are shared between booth tasks. Then the output of hte 6 the layer is spit up into twp separate tranformers treams , one for modes, and one for deinitealerving. We use 2 lauers each here.

the output is the finally also tranofrmaed using a projection head for each from d-modeel to 16 dims for modes clustering and 8 for deiniteralvng.

D model is 256 for the shared layers, with 8 heads and 2048 dim deedforamweds.

The splitted tranfomrer blocks wach have 2 layers, d model is 128, 4 heads and dim feedforward sis 1024.

4.4 Loss Function and Clustering Objective

Placeholder paragraph.

4.5 Training Procedure

Placeholder paragraph.

4.6 Evaluation Metrics

Placeholder paragraph.

4.7 Baseline Methods

Placeholder paragraph.

CHAPTER 5

Experiments and Results

5.1 Experimental Setup

Placeholder paragraph.

5.2 Datasets

Placeholder paragraph.

5.3 Hyperparameters and Training Details

Placeholder paragraph.

5.4 Quantitative Results

Placeholder paragraph.

5.5 Ablation Studies

Placeholder paragraph.

5.6 Qualitative Analysis

Placeholder paragraph.

CHAPTER 6

Discussion

6.1 Interpretation of Results

Placeholder paragraph.

6.2 Comparison with Related Work

Placeholder paragraph.

6.3 Limitations

Placeholder paragraph.

6.4 Implications

Placeholder paragraph.

CHAPTER 7

Conclusion

7.1 Summary

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7.2 Answers to the Research Questions

Placeholder paragraph.

7.3 Future Work

Placeholder paragraph.

APPENDIX A

Additional Results

Placeholder paragraph.

APPENDIX B

Implementation Details

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APPENDIX C

Notation Reference

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